Progress Report: Cross-Regime Generalization in Cryptocurrency Direction Prediction

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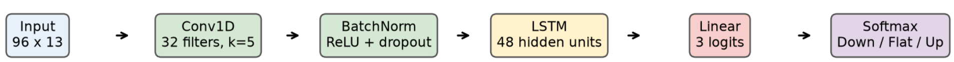
Google Colab notebook

1 Brief Project Description

Cryptocurrency forecasting is difficult because the market is continuous, volatile, nonlinear, and non-stationary. This project studies not only whether recent market history predicts the next hourly direction of BTC and ETH, but whether any learned signal remains reliable when the market regime changes. Each input is a 96×13 tensor: 96 hourly observations described by returns, price ranges, volume changes, RSI, MACD, Bollinger features, and rolling volatility. The output is one of three classes—*down*, *flat*, or *up*—defined from the next-hour close return.

Deep learning is reasonable because useful evidence may occur as short local motifs and as longer temporal dependencies. The 1D convolution searches each window for local patterns, while an LSTM summarizes their order and longer context. Prior work has found value in LSTMs for financial direction prediction, while also reporting deterioration across sub-periods [1]; hybrid CNN–LSTM models have likewise been applied to Bitcoin [2]. This flexibility makes the model a useful subject for the eventual cross-regime stress test, even if its first checkpoint result is modest.



2 Notable Contribution

2.1 Data Processing

We downloaded 26,304 hourly Binance klines for each of BTCUSDT and ETHUSDT from July 1, 2023 to June 30, 2026 (52,608 rows total). Rows were parsed to UTC, sorted, deduplicated, and validated for numeric OHLCV values. We computed 13 causal features: one-hour simple/log returns, high–low range, close–open return, volume change, RSI(14), MACD(12,26,9), Bollinger position/width(20), 24-hour volatility, and 24-hour volume z-score. After rolling warm-up, 26,280 clean feature rows remained per asset. A training-only median absolute next-hour return threshold (0.2357%) defined down/flat/up.

Splits are chronological (70/15/15), windows never cross assets or split boundaries, and z-score parameters are fitted only on usable training rows. This produced 36,584 training, 7,698 validation, and 7,702 test windows. The main challenges were paginated API collection, class imbalance, and preventing look-ahead leakage at feature, label, scaling, and window boundaries. Figure 1 shows the cleaned series and class statistics. For the final report, hourly observations collected **after July 1, 2026** will be locked as never-before-seen data and **not used for tuning**.

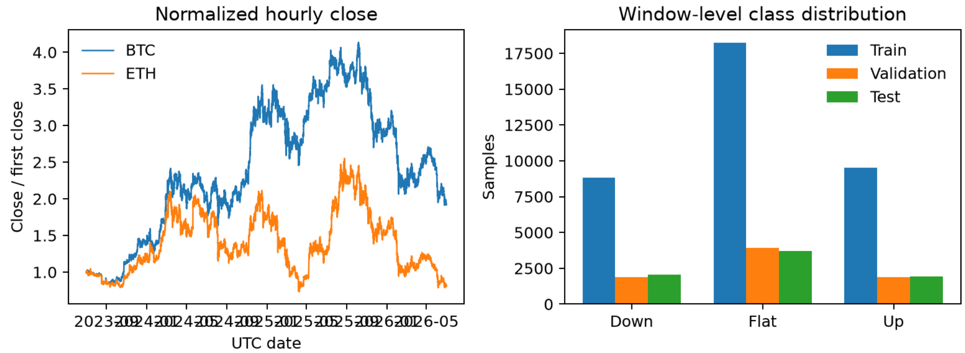


Figure 1: Clean hourly price paths and window-level class counts. The test set contains 2,061 down, 3,703 flat, and 1,938 up examples.

2.2 Baseline Model

Two deliberately simple, memory-free baselines measure whether sequence modeling adds value. The momentum heuristic maps the most recent observed return through the same training-derived class threshold. Logistic regression uses the final 13-feature vector, balanced class weights, and otherwise default regularization. Both models are fitted without test-set tuning and evaluated on exactly the same 7,702 chronological test windows as the CNN–LSTM.

Table 1: Held-out next-hour classification results. Recall order is down/flat/up.

Model	Accuracy	Macro-F1	Recall D	Recall F	Recall U
Momentum	0.421	0.380	0.295	0.564	0.281
Logistic regression	**0.480**	**0.429**	0.271	**0.656**	0.366
CNN–LSTM	0.444	0.425	**0.403**	0.498	**0.386**

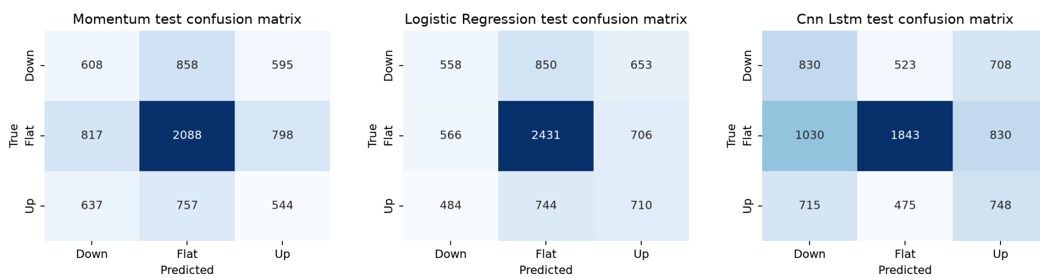


Figure 2: Test confusion matrices for momentum, logistic regression, and CNN–LSTM (left to right).

Quantitative and qualitative result. Logistic regression is the strongest checkpoint by accuracy (48.0%), improving by 5.9 percentage points over momentum. Figure 2 reveals why: it recognizes 65.6% of flat examples but only 27.1% of down examples. Thus the apparently stronger aggregate score is concentrated in the largest class rather than uniformly reliable directional prediction. This is a useful baseline for the primary model because it is cheap, reproducible, and exposes whether temporal processing improves minority-direction recall.

Challenge. The classes remain imbalanced even with a training-only, data-dependent threshold. Accuracy alone can therefore reward predicting flat; macro-F1 and per-class recall are retained as primary companion metrics. No metric here is interpreted as profitability.

2.3 Primary Model

The PyTorch model preserves the proposal's hybrid design. A same-padded Conv1D maps 13 features to 32 channels with kernel size 5, followed by ReLU, batch normalization, and 0.15 dropout. A single LSTM with 48 hidden units processes the 96 enriched timesteps; the final hidden output passes through 0.20 dropout and a $48 \rightarrow 3$ linear head. The model has 18067 trainable parameters. We use weighted cross-entropy, Adam (10^{-3}), batch size 256, gradient clipping at 1.0, and a fixed seed. The maximum is 12 epochs; validation-loss patience of three selected epoch 4 and stopped after epoch 7.

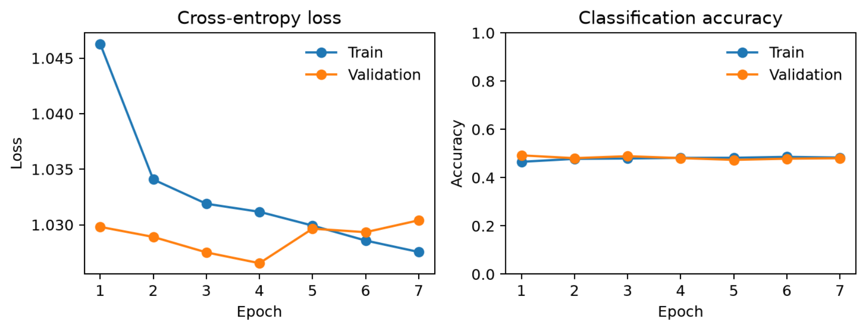


Figure 3: Training and validation curves. Validation loss improved from 1.030 to 1.027 by epoch 4, establishing that the full neural pipeline trains, before mild divergence triggered early stopping.

Quantitative result. The CNN–LSTM reaches 0.444 accuracy and 0.425 macro-F1. It **did not outperform** logistic regression on accuracy, but improves down recall from 0.271 to 0.403 and up recall from 0.366 to 0.386. This suggests that temporal context trades some majority-class performance for more balanced directional sensitivity; further experiments must determine whether that trade-off survives regime shift.

Qualitative result and challenge. The model correctly labels a BTC flat case on April 4, 2026 with 86.6% confidence, but also labels an actual up case on May 2 as flat with 85.9% confidence. The most confident correct and error examples are dominated by flat predictions, consistent with the 48.1% flat share of the test set. Weighted loss helps minority recall but does not remove overconfidence or non-stationarity.

Feasibility and next steps. The checkpoint demonstrates a reproducible end-to-end pipeline, two useful baselines, successful neural optimization, and held-out quantitative/qualitative evaluation. For the final project we will freeze the new post-July data, define volatility regimes, test cross-regime degradation, and tune only on validation data. The current result is a feasibility study, **not trading advice**, and no profitability claim is made.

References

[1] Thomas Fischer and Christopher Krauss. Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654–669, 2018.

[2] Qiutong Guo, Shun Lei, Qing Ye, and Zhiyang Fang. MRC-LSTM: A hybrid approach of multi-scale residual CNN and LSTM to predict bitcoin price. *arXiv preprint arXiv:2105.00707*, 2021.